Quiz 2: MA 5104, Department of Mathematics, IIT Madras

Name : Statistical Inference and Learning	**Maximum Marks: 100**

Roll no: ID25S027	Name: PUNEET	A

- The exam duration is 50 minutes.
- Write **ALL Formulas/Theorems/Results** used.
- No need to write the entire question on your answer sheets.
- Write your roll number and name on both the question paper and the answer sheet.
- Calculators are allowed, but sharing them is NOT allowed.
- You do not need to return the question paper. Only the answer sheet will be collected.
- If the answer is a rational number, leave it as a fraction or in decimals up to 3 decimal places.

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1. (**20 marks**) Write down ONE correct answer for each of the sub-parts:

   (i) Which of the following is *not* a legitimate null hypothesis?

   (A) $H_0 : \sigma^2 = 4$

   (B) $H_0 : \mu_1 - \mu_2 = 0$

   (C) $H_0 : \bar{x} = 12.5$

   (D) $H_0 : p \leq 0.3$

   (ii) Holding the sample size and the true effect fixed, lowering the significance level from $\alpha = 0.05$ to $\alpha = 0.01$

   (A) decreases the probability of a Type II error.

   (B) increases the probability of a Type II error.

   (C) leaves the power of the test unchanged.

(D) decreases both error probabilities simultaneously.

(iii) A sample of $n = 12$ observations is drawn from a normal population with unknown variance, and we wish to test $H_0 : \mu = \mu_0$. The appropriate reference distribution is

(A) the standard normal.

(B) t with 12 degrees of freedom.

(C) t with 11 degrees of freedom.

(D) χ^2 with 11 degrees of freedom.

(iv) A sample of $n = 400$ is drawn from a strongly right-skewed population and we test a hypothesis about the mean. The best justification for using a normal-based test is that

(A) skewness never affects any hypothesis test.

(B) the central limit theorem makes the sampling distribution of $\bar{X}$ approximately normal.

(C) a large sample guarantees that the individual observations are approximately normal.

(D) the test is exact for any population provided $n > 30$.

(v) The χ^2 approximation to the null distribution of the test statistic is unreliable when

(A) several expected cell counts are small.

(B) several observed cell counts are small but the expected counts are large.

(C) the total sample size exceeds 1000.

(D) the table has equal numbers of rows and columns.

(vi) In the nonparametric bootstrap, each resample is drawn

(A) with replacement from the observed sample, and has size n.

(B) without replacement from the observed sample, and has size n.

(C) from a normal distribution fitted to the observed sample.

(D) directly from the population.

(vii) A 95% credible interval for θ is reported as $(2.1,\ 7.9)$. The correct interpretation is

(A) $P(\theta \in (2.1, 7.9) \mid \text{data}) = 0.95$.

(B) in repeated sampling, 95% of such intervals would contain θ.

(C) 95% of the observed data lie between 2.1 and 7.9.

(D) there is a 5% chance of a Type I error if θ lies outside the interval.

(viii) Which statement about credible intervals is correct?

(A) A credible interval does not depend on the choice of prior.

(B) With a flat prior and a normal likelihood, a credible interval can coincide numerically with the corresponding confidence interval, even though the two are interpreted differently.

(C) A credible interval is always narrower than the corresponding confidence interval.

(D) Credible intervals are valid only for large samples.

2. (**10 marks**) A buyer receives a shipment of components from a supplier. Before paying for it and releasing it to the production line, the buyer's quality inspector draws a sample and decides whether to accept the shipment or return it to the supplier. On this occasion the inspector accepts it.

(a) What hypotheses are being tested if accepting the shipment is a type II error?

(b) Under those hypotheses, which party (the buyer or the supplier) bears the risk controlled by α? Explain in one sentence.

3. The following data represent the length of time, in days, to recovery for patients randomly treated with one of two medications to clear up severe bladder infections:

$$\text{Medication 1} \quad n_1 = 14 \quad \bar{x}_1 = 17 \quad s_1^2 = 1.5$$
$$\text{Medication 2} \quad n_2 = 16 \quad \bar{x}_2 = 19 \quad s_2^2 = 1.8$$

(i) **(18 marks)** Test, at the 1% level of significance, whether the two medications differ in mean recovery time. Assume the two populations are normal with equal variances. Note that $t_{0.005,28} = 2.763$ and $t_{0.01,28} = 2.467$.

Your answer must show each of the following steps explicitly, and will be marked on all of them:

(a) State the parameter of interest and the null and alternative hypotheses.

(b) State the test statistic and its distribution under H_0, including the degrees of freedom, and state the assumptions required for that distribution to hold.

(c) State the rejection region for $\alpha = 0.01$.

(d) Compute the pooled variance s_p^2 and the observed value of the test statistic. 1.6607 -4.241

(e) State your decision. reject H_0

(f) State your conclusion in the context of the problem, in one sentence, avoiding the word "significant".

(ii) (**8 marks**) Find a 99% confidence interval for $\mu_1 - \mu_2$. State how the interval could have been used to reach the decision in (a) without computing the test statistic, and explain in one sentence why the two approaches must agree.

(iii) (**10 marks**) A clinician argues that medication 1 was already known to act faster, so the alternative should have been $H_1 : \mu_1 < \mu_2$. Redo the test under this alternative, stating clearly:

(a) the test statistic and its distribution

(b) the rejection region;

(c) which of the quantities computed in (i) change, and which do not;

(d) the decision;

(e) one condition that must be satisfied for this choice of alternative (that is, the one-sided) to be legitimate.

4 The Department of Mathematics runs two postgraduate programmes: the **M.Sc. (Mathematics)** and the **M.Tech. (Industrial Mathematics)**. In the most recent placement season, students applied to roles of two kinds: **Analytics** profiles, of which there were many openings, and **Quantitative Finance** profiles, of which there were very few. Each student applied to exactly one kind of role. An M.Sc. student representative petitions the Placement Committee, claiming that recruiters on this campus systematically favour M.Tech. candidates. She submits the following table.

All applicants combined

	Offer	No offer	Total
M.Tech.	172	78	250
M.Sc.	98	152	250
Total	270	230	500

The Committee asks for the same counts broken down by the kind of role.

Analytics profiles			**Quantitative Finance profiles**		
	Offer	No offer		Offer	No offer
M.Tech.	160	40	M.Tech.	12	38
M.Sc.	42	8	M.Sc.	56	144

You may use $\chi^2_{0.05,1} = 3.841$ and $\chi^2_{0.01,1} = 6.635$.

(a) (**12 marks**) Using the combined table, test at the 5% level whether programme and placement outcome are independent. State the hypotheses, the expected counts, the test statistic and its null distribution with degrees of freedom, the rejection region, your decision, and your conclusion in context.

(b) (**15 marks**) Repeat the test separately for Analytics profiles and for Quantitative Finance profiles. Report χ^2, the degrees of freedom, and your decision in each case.

(c) (**7 marks**) Your answers to (a) and (b) point in opposite directions, and both calculations are correct. Explain how this is possible, and state which analysis the Placement Committee should act on, with justification.